Isoelectric Point — is a protein acidic?

Data Analysis & Visualization — labs 11-13

Binary classification of proteins and peptides into acidic (pI < 5, label 1) and non-acidic (label 0). This report covers the four parts of the project: building a balanced dataset from the IPC protein and peptide FASTA files, engineering sequence features, exploring them, training classical ML models, and finally a dense neural network with a command-line inference tool.

Part 1 — Dataset & exploration

```
Parsed sources:  proteins = 2183,  peptides = 16882

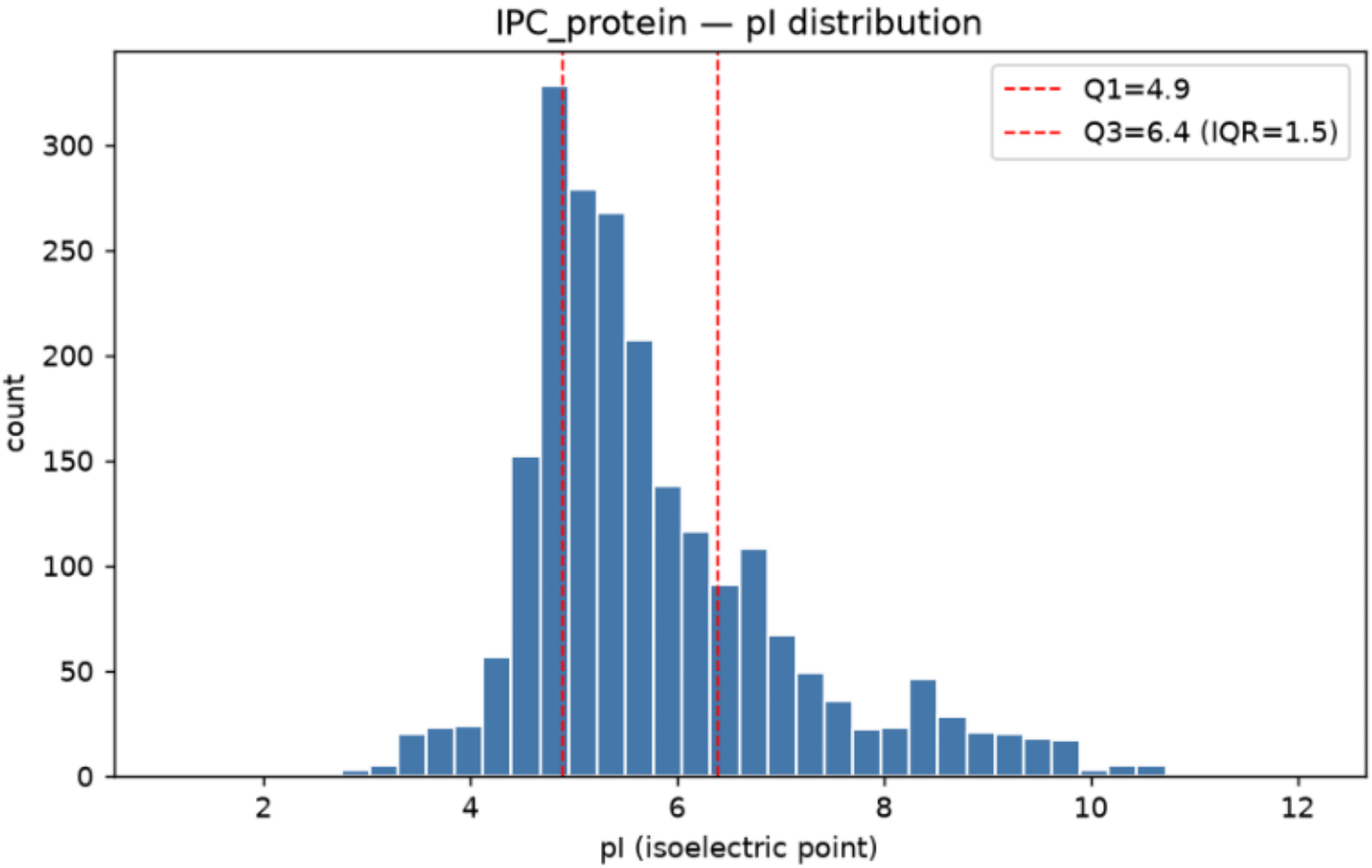
Balanced classification dataset (2200 items, 1100 acidic / 1100 non-acidic):
  550 protein positives  (pI < 5)
  550 protein negatives  (the 550 most basic proteins)
  550 peptide positives  (random, pI < 5)
  550 peptide negatives  (random, pI > 10)

UID = MD5(sequence). Header format:  >UID|type|pI|label
Split 60/20/20 two ways: plain random, and stratified by source type.
The stratified split keeps proteins/peptides balanced across train/test/val.

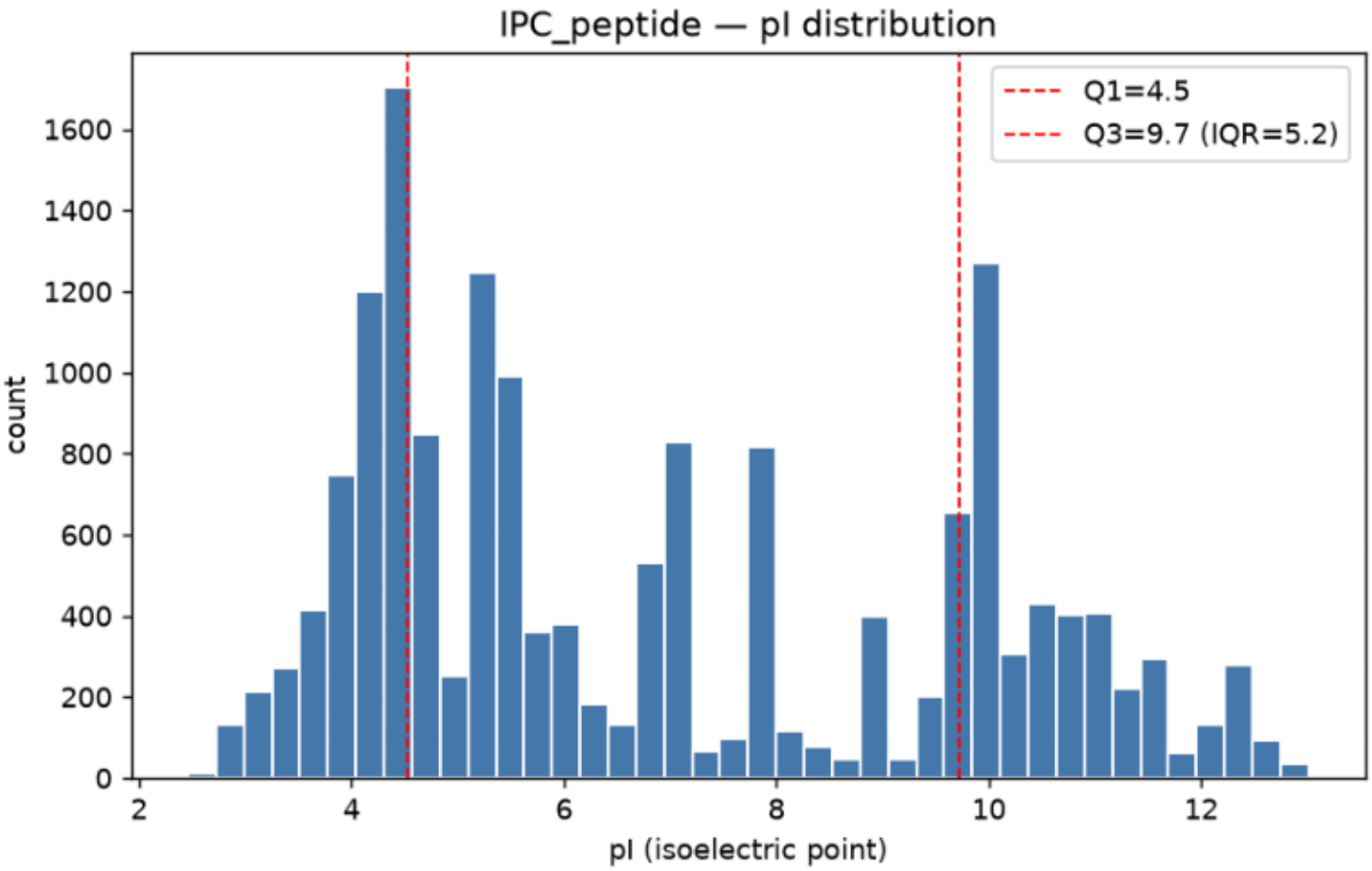
Interquartile ranges:
IPC_protein  pI      Q1=    4.90  Q3=    6.40  IQR=    1.50
IPC_protein  length  Q1=  221.50  Q3=  494.00  IQR=  272.50
IPC_peptide  pI      Q1=    4.53  Q3=    9.72  IQR=    5.19
IPC_peptide  length  Q1=   12.00  Q3=   20.00  IQR=    8.00
```

Part 1 — pI & length distributions

Protein pI (Q1/Q3 marked)

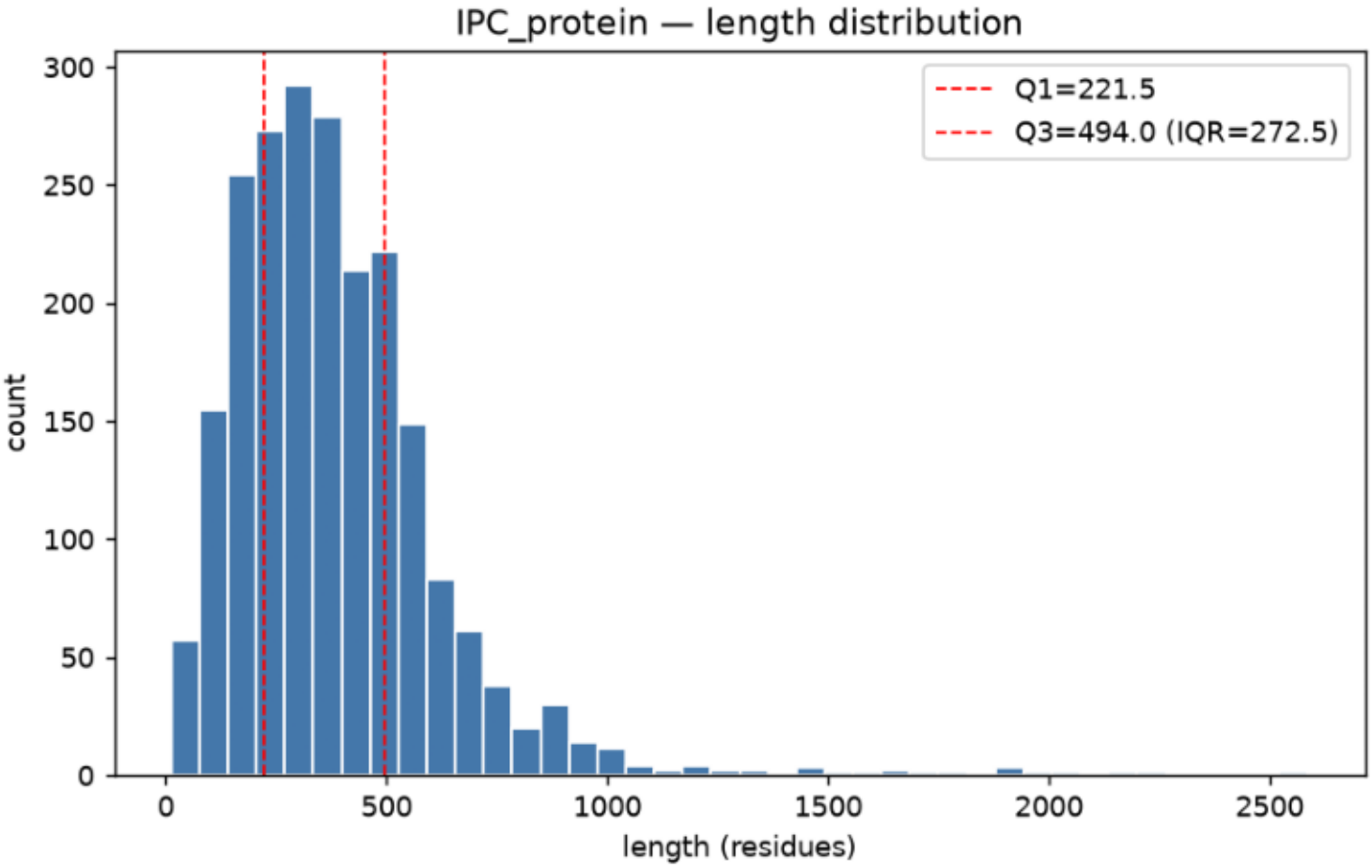


Peptide pI (Q1/Q3 marked)

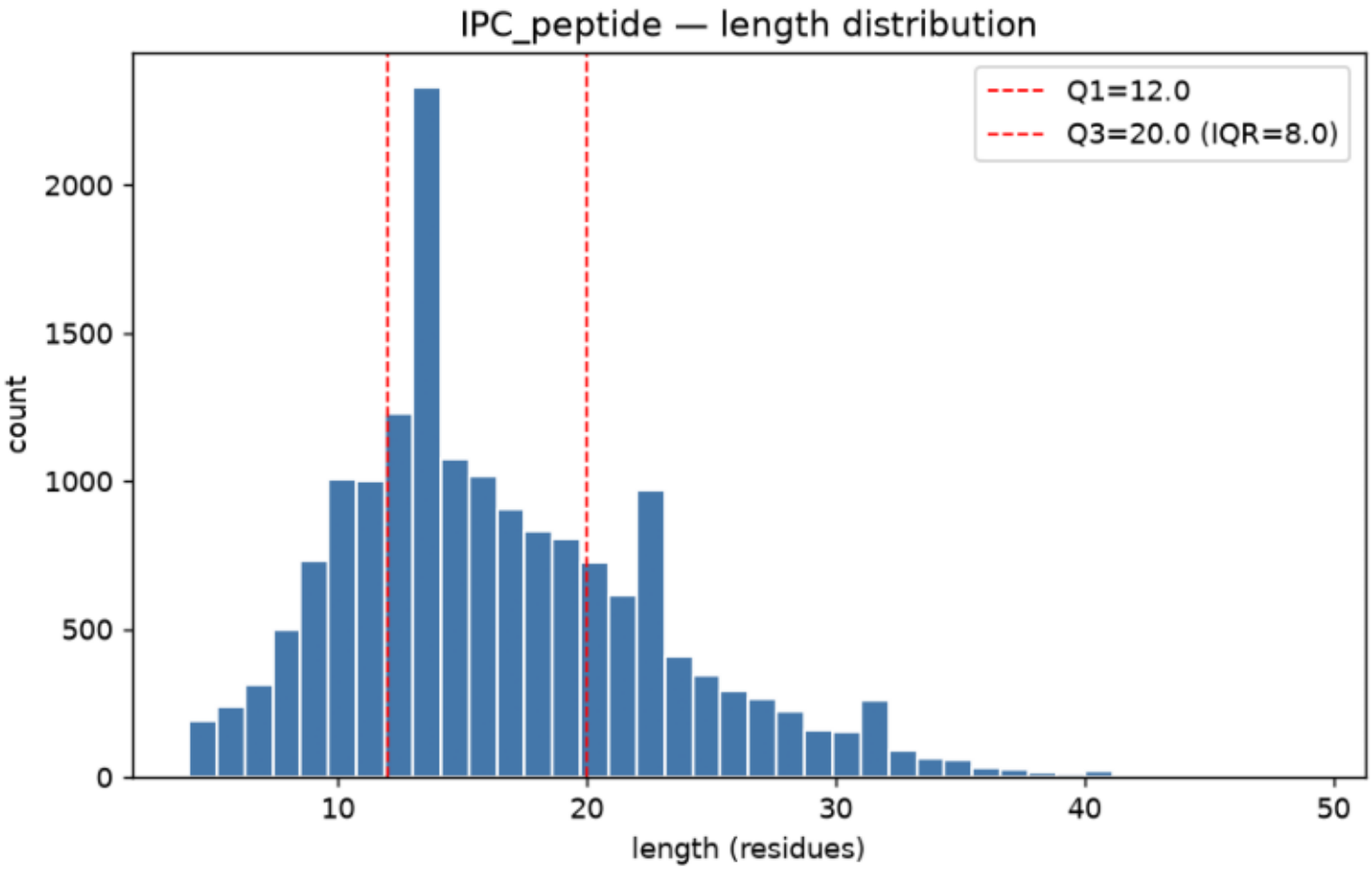


Part 1 — sequence length distributions

Protein length



Peptide length



Part 2 — Features & classical models

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Features (sequence-derived only; pI is excluded to avoid label leakage):
  length, molecular_weight, 20 amino-acid counts, hydrophobicity
  (Kyte-Doolittle total & mean), net charge.

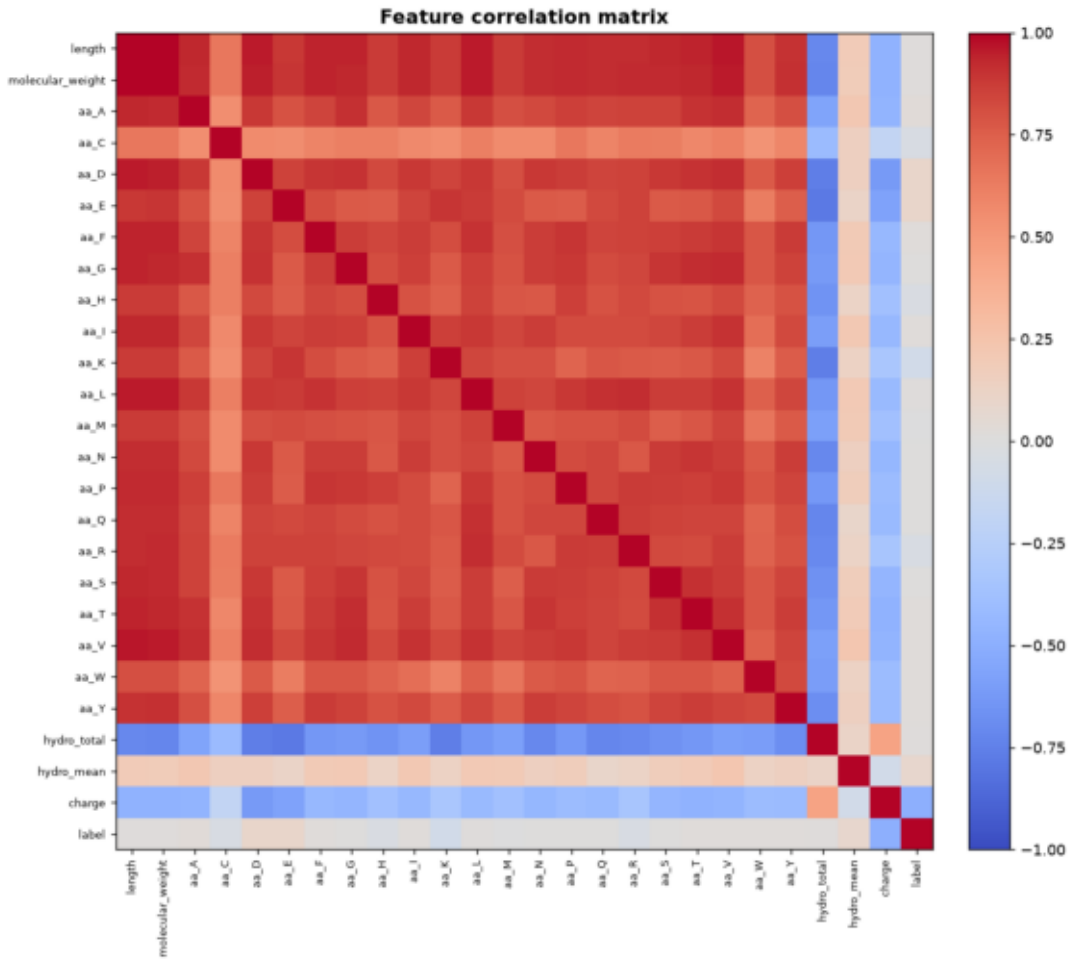
The strongest single predictor is net charge (Pearson r = -0.50 with the
acidic label): acidic sequences carry more D/E and fewer K/R. The acidic
residue counts aa_D and aa_E are the next most informative.

Held-out test accuracy:

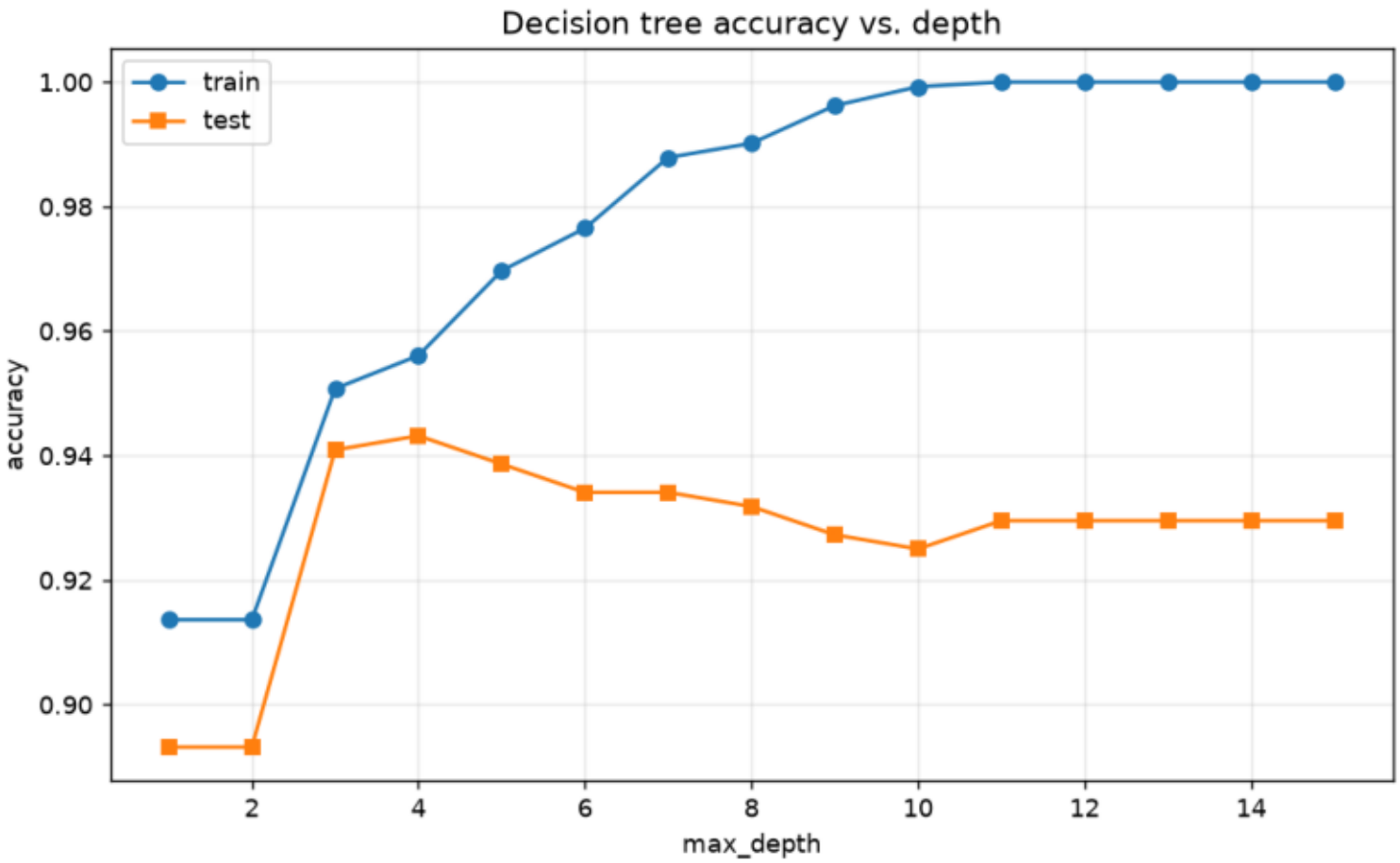
       model                                best_params  test_accuracy
DecisionTree                                max_depth=4         0.9432
RandomForest {'max_depth': None, 'n_estimators': 300}         0.9386
         KNN {'kneighborsclassifier__n_neighbors': 5}         0.9205
   SVM (RBF)       {'svc__C': 10, 'svc__gamma': 0.01}         0.9477
```

Part 2 — feature correlations & tree tuning

Feature correlation matrix

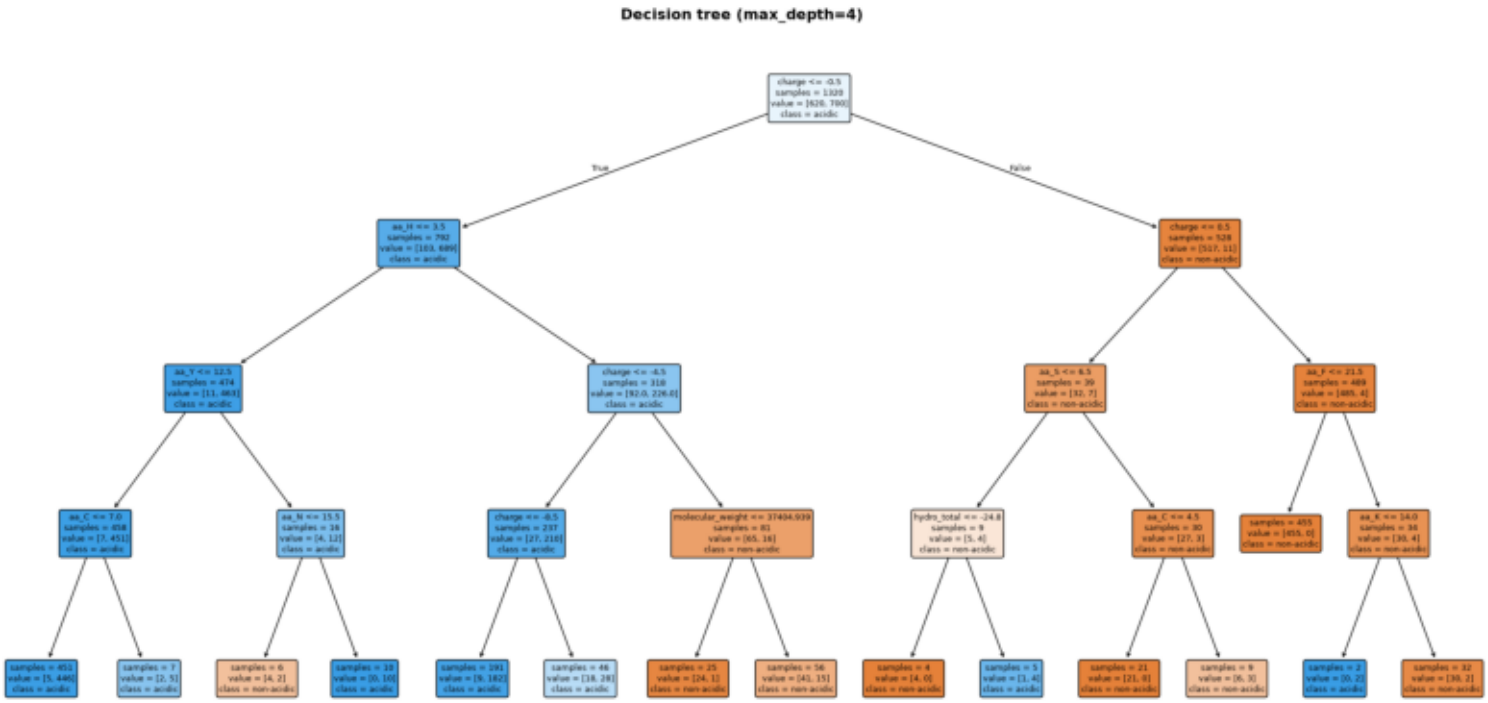


Decision-tree accuracy vs depth



Part 2 — decision tree (max_depth = 4)

Charge and the acidic-residue counts dominate the top splits



Parts 3-4 — Dense neural network & inference

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Model: scikit-learn MLPClassifier (dense), relu + adam, L2 alpha=1e-3,
early stopping (dropout-style regularisation).

Architecture (in -> hidden -> hidden -> out):
  layer 1: 25 -> 64
  layer 2: 64 -> 32
  layer 3: 32 -> 1

Converged in 28 iterations.
Test accuracy: 0.9341

Saved in three formats: acidic_mlp.joblib (full pipeline),
acidic_mlp.json (architecture summary), acidic_mlp_weights.npz (weights).

Inference tool (acidic_protein_dl_predictor.py) decorates FASTA headers:
  >protein 1 | acidic 0.917
  >UPI00028EB81A status=active | non-acidic 0.909
  >AF-A0A782FH33 | non-acidic 0.944

Conclusion: all models reach ~92-95% accuracy on this balanced task;
the SVM (RBF) and decision tree lead marginally. Net charge and acidic
residue composition carry most of the signal, which matches the biochemistry
of the isoelectric point.
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